

LAN8650/1 Configuration Application Note

INTRODUCTION

This application note provides guidance on how to configure the LAN8650/1 for optimal performance in 10BASE-T1S networks. Microchip is constantly developing improvements to the configuration. For the most current specifications, please visit the product pages on the website for [LAN8650 and LAN8651 MAC-PHY Ethernet Controller with SPI \[1\]](#).

The information presented in this application note is based upon the LAN8650/1 hardware revisions listed in the following table and is applicable for these and newer versions of the device until superseded by a newer version of the Configuration Application Note.

Product Revision ¹	Silicon Revision ²	Data Sheet ³
B0	0001	DS60001734
B1	0010	DS60001734

Note 1: The product revision is noted in the package top marking.

2: The silicon revision is obtained by reading the Revision field from the Device Identification (DEVID) register.

3: All specifications can be found in the latest revision of the device data sheet.

ACCESSING CONFIGURATION REGISTERS

The configuration settings for the LAN8650/1 consist of a series of configuration register accesses performed through the Serial Peripheral Interface (SPI). The register accesses must be written immediately following a power cycle or reset of the device and should be performed in the sequence provided in this application note. Details for accessing the configuration registers through the SPI may be found in the device data sheet.

The types of register functions utilized in the configuration include:

- W – Write of value to specified register
- R – Read of value from specified register
- IR - Indirect read of value from specified register

Read Pseudocode

```
(uint16) value = read_register(uint8 mms, uint16 addr)
```

Write Pseudocode

```
write_register(uint8 mms, uint16 addr, uint16 value)
```

where:

mms	Memory Map Selector
address	Address of the register to access
value	Value to be written to or read from the register at address

Indirect Read Pseudocode

Proprietary access mechanism. Do not confuse this with the Clause 22 indirect access to Clause 45 registers.

```
uint8 indirect_read(uint8 addr, uint8 mask)
{
    write_register(0x04, 0x00D8, addr)
    write_register(0x04, 0x00DA, 0x2)
    return (read_register(0x04, 0x00D9) & mask)
}
```

CONFIGURATION PROCESS

Before configuring the LAN8650/1, some device configurations must be read and saved for proper configuration of the device.

Calculation of configuration parameters

Value1 has an expected range from -5 to 15. Value1 is stored as a signed 5-bit value in address location 0x0004.

```
uint8 value1 = indirect_read(0x04,0x1F)
int8 offset1
if((value1 & 0x10) != 0)
{
    offset1 = (int8) ((uint8)value1 - 0x20)
}
else
{
    offset1 = (int8) value1
}
```

Value2 is stored as a signed 5-bit variable in address location 0x0008.

```
uint8 value2 = indirect_read(0x08,0x1F)
int8 offset2
if((value2 & 0x10) != 0)
{
    offset2 = (int8) ((uint8)value2 - 0x20)
}
else
{
    offset2 = (int8) value2
}
```

The following parameters must be calculated from the device configuration parameters mentioned above to use for the configuration of the registers.

```
uint16 cfgparam1 = (uint16) (((9 + offset1) & 0x3F) << 10) | (uint16) (((14 + offset1) & 0x3F) << 4) | 0x03
uint16 cfgparam2 = (uint16) (((40 + offset2) & 0x3F) << 10)
```

Write Configuration into Registers

Table 1 lists a series of parameters that must be configured with the given values and in the order listed. In addition to configuring the PHY transceiver in the device, the following configuration configures the MAC to set time stamping at the end of the Start of Frame Delimiter (SFD) and set the Timer Increment register to 40 ns to be used as a 25 MHz internal clock.

TABLE 1: CONFIGURATION REGISTER WRITES

Access	MMS	Address	Value
W	0x4	0x00D0	0x3F31
W	0x4	0x00E0	0xC000
W	0x4	0x0084	cfgparam1
W	0x4	0x008A	cfgparam2
W	0x4	0x00E9	0x9E50
W	0x4	0x00F5	0x1CF8
W	0x4	0x00F4	0xC020
W	0x4	0x00F8	0xB900
W	0x4	0x00F9	0x4E53
W	0x4	0x0081	0x0080

TABLE 1: CONFIGURATION REGISTER WRITES (CONTINUED)

Access	MMS	Address	Value
W	0x4	0x0091	0x9660
W	0x1	0x0077	0x0028
W	0x4	0x0043	0x00FF
W	0x4	0x0044	0xFFFF
W	0x4	0x0045	0x0000
W	0x4	0x0053	0x00FF
W	0x4	0x0054	0xFFFF
W	0x4	0x0055	0x0000
W	0x4	0x0040	0x0002
W	0x4	0x0050	0x0002

(Optional) Write SQI Configuration into Registers

If the SQI feature is to be used, the following configuration parameters must be calculated and saved so that they can be written into the corresponding registers in the next step.

```
uint16 cfgparam3 = (uint16) (((5 + offset1) & 0x3F) << 8) | (uint16) ((9 + offset1) & 0x3F)
uint16 cfgparam4 = (uint16) (((9 + offset1) & 0x3F) << 8) | (uint16) ((14 + offset1) & 0x3F)
uint16 cfgparam5 = (uint16) (((17 + offset1) & 0x3F) << 8) | (uint16) ((22 + offset1) & 0x3F)
```

Table 2 contains the register elements that must be configured with the parameters listed and in the order given.

TABLE 2: SQI CONFIGURATION REGISTER WRITES

Access	MMS	Address	Value
W	0x4	0x00AD	cfgparam3
W	0x4	0x00AE	cfgparam4
W	0x4	0x00AF	cfgparam5
W	0x4	0x00B0	0x0103
W	0x4	0x00B1	0x0910
W	0x4	0x00B2	0x1D26
W	0x4	0x00B3	0x002A
W	0x4	0x00B4	0x0103
W	0x4	0x00B5	0x070D
W	0x4	0x00B6	0x1720
W	0x4	0x00B7	0x0027
W	0x4	0x00B8	0x0509
W	0x4	0x00B9	0x0E13
W	0x4	0x00BA	0x1C25
W	0x4	0x00BB	0x002B

PLCA Configuration Process

Under normal operation, the device should be operated in PLCA mode. Disabling collision detection is recommended to allow the device to operate in noisy environments or when reflections and other inherent transmission line distortion cause poor signal quality. Disabling collision detection is also recommended during initial bring-up and debug. If the device determines that PLCA mode is not possible, it will revert into a fallback mode where PLCA is disabled and communication is handled in CSMA/CD mode. In this case, collision detection must be re-enabled.

Enabling PLCA

When using PLCA, the node with the Node_ID **0** is called the PLCA coordinator. For the PLCA coordinator, the number of transmit opportunities, which is the maximum Node_ID used in the mixing segment plus 1, must be written into the PLCA Control 1 (PLCA_CTRL1, Address = 0xCA02) register.

Nodes with IDs ranging between 1 and 0xFE are called PLCA followers. These nodes only require their Node_ID to be written into the field of PLCA_CTRL1.

Once the number of transmit opportunities and Node_ID have been written, enable PLCA by writing a '1' into the PLCA Enable (EN) bit in the PLCA Control 0 (PLCA_CTRL0, Address = 0xCA01) register.

The following code configures the device as either a PLCA coordinator or PLCA follower depending on the device's corresponding Node ID.

```
uint8 Node_ID
uint8 Node_Count
uint16 plcaparam1

if(Node_ID == 0)
{
    plcaparam1 = (uint16) (Node_Count << 8)
}
else
{
    plcaparam1 = (uint16) Node_ID
}
```

In many cases, as described above, collision detection should be disabled by writing a '0' into the Collision Detection Enable (CDEN) bit of the Collision Detection Control 0 (CDCTL0, Address = 0x0087) register.

CAUTION: When writing to CDCTL0, use a read-modify-write operation to avoid accidental modifications to reserved fields.

This sequence is summarized in [Table 3](#).

TABLE 3: PLCA CONFIGURATION REGISTER WRITES

Access	MMS	Address	Value
W	0x4	PLCA_CTRL1	plcaparam1
W	0x4	PLCA_CTRL0	0x8000
R	0x4	CDCTL0	CDCTL0_TEMP
W	0x4	CDCTL0	CDCTL0_TEMP & (~0x8000)

Managing Collision Detection

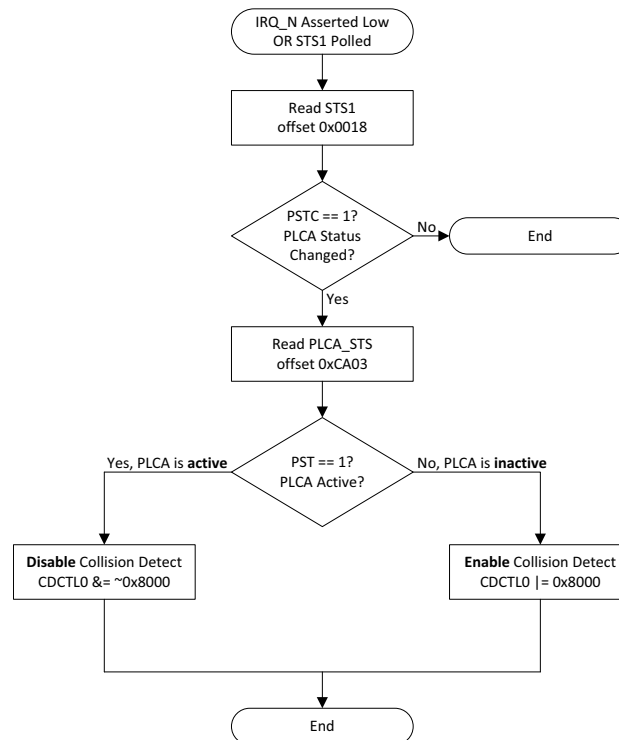
When configured for PLCA operation, it may be advantageous to disable collision detection. Should the device not detect the periodic presence of a PLCA BEACON, it will automatically fall back to CSMA/CD operation to keep communications running. Whenever PLCA is active, the PLCA Status (PST) bit in the PLCA Status (PLCA_STS, Address = 0xCA03) register is set by hardware and changes from a '0' to a '1'. Similarly, whenever the PLCA BEACON is lost and the device falls back to CSMA/CD operation, the bit changes from a '1' to a '0'. Whenever these changes occur, a '1' is set in the PLCA Status Changed (PSTC) bit in the Status 1 (STS1, Address = 0x0018) register. This flag is used to indicate that the activity status of PLCA has changed since the last read of the register.

When disabling collision detection while PLCA is active, it is necessary to manually re-enable it during the CSMA/CD fallback. The best way to do this is to enable an interrupt to notify the microcontroller whenever a PLCA status change has occurred. This is done by writing a '0' into the PLCA Status Changed Mask (PSTCM) bit in the Interrupt Mask 1 (IMSK1, Address = 0x001C) register. Alternatively, if no interrupts are being used, the device can be polled regularly. This can, however, lead to reduced performance.

Whenever an interrupt is triggered through the IRQ_N pin or a poll occurs, the STS1 register should be read. If the PLCA Status Changed (PSTC) bit is set, then collision detection should be enabled or disabled as shown in [Figure 1](#). During the handling of this interrupt or poll, the PLCA_STS register will indicate if PLCA is currently active or not. If PLCA is

currently active, as indicated by reading the PLCA Status bit as '1', collision detection is disabled by clearing the Collision Detection Enable (CDEN) bit in the Collision Detect Control 0 (CDCTL0, Address = 0x0087) register. When the PLCA Status bit is read as '0', indicating that PLCA is inactive and the device is operating in CSMA/CD, then collision detection is enabled by writing a '1' to the CDEN bit.

FIGURE 1: PLCA COLLISION DETECT FLOWCHART



Disabling PLCA

In some cases, it may be desired to force the device to remain in CSMA/CD mode. To do this, disable PLCA in the PLCA_CTRL0 register and enable collision detect in the CDCTL0 register.

TABLE 4: RETURN TO CSMA/CD FROM PLCA

Access	MMS	Address	Value
R	0x04	CDCTL0	CDCTL0_TEMP
W	0x04	CDCTL0	CDCTL0_TEMP (0x8000)
W	0x04	PLCA_CTRL0	0x0000

APPENDIX A: REFERENCES

This application note is intended to be used in conjunction with the following products and documents:

- [1] LAN8650 and LAN8651 MAC-PHY Ethernet Controller with SPI
<https://www.microchip.com/LAN8650>
<https://www.microchip.com/LAN8651>
- [2] LAN8650/1: 10BASE-T1S MAC-PHY Ethernet Controller with SPI Data Sheet (60001734)
<https://www.microchip.com/LAN8650>

APPENDIX B: APPLICATION NOTE REVISION HISTORY

B.1 Current Document Revision

Revision F (DS60001760G - June 2024)

- Updated register configuration process
- Clarified calculation of configuration parameters

B.2 Previous Document Revision

Revision E (DS60001760E - November 2023)

- Updated configuration register writes to clarify SQL configuration
- Updated PLCA and CSMA/CD configuration

Revision D (DS60001760D - July 2023)

- Updated to reflect configurations for Product Revision B1.
- Updated Table 1, Configuration Register Reads.
- Updated Table 2, Configuration Register Writes.

Revision C (DS60001760C - July 2023)

- Updated Table 1, Configuration Register Reads.

Revision B (DS60001760B - February 2023)

- Updated to reflect configurations for Product Revision B0.

Revision A (DS60001760A - March 2022)

- Initial Release to reflect configurations for Product Revision A0.

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